OPTICS: Automatic MT Evaluation Based on Optimal Transport by Integration of Contextual Representations and Static Word Embeddings

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Abstract. Automatic MT metrics using word embeddings are extremely effective. Semantic word similarities are obtained using word embeddings. However, similarities using only static word embeddings are insufficient for lack of contextual information. Automatic metrics using fine-tuned models can adapt to a specific domain using contextual representations obtained by learning, but that adaptation requires large amounts of data to learn suitable models. As described herein, we propose an automatic MT metric based on optimal transport using both contextual representations and static word embeddings. The contextual representations are obtained by learning the neural models. In that case, our proposed metric requires no other data except source sentences and references, which correspond to the evaluation target hypotheses, to learn the models that are used to extract the contextual representations. Therefore, our proposed metric can adapt to the domain appropriately without requiring large amounts of learning data. Experiment results obtained using the WMT 20 metric shared task data indicated that correlations with human judgment using our proposed metric are higher than those using a metric based only on static word embeddings. Moreover, our proposed metric achieved state-of-the-art performance with system-level correlation and to-English segment-level correlation.

Keywords: Automatic metric · Contextual representations · Optimal transport · Static word embeddings

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1 Introduction

Automatic metrics are necessary for machine translation research. Quantitative evaluation of machine translation systems without an automatic metric is difficult because human evaluation needs expensive and time-consuming. Therefore, the metrics which can achieve high correlation with human judgment are necessary, and the investigation for the metrics, that have the highest accuracy, is also performed [1]. Actually, BLEU [2] has been used widely as an industry standard metric to evaluate various machine translation systems. Nevertheless, metrics based on surface matching (*e.g.* BLEU, TER [3], chrF [4], chrF++ [5]) are unsuitable to measure semantic similarity between a reference and hypothesis because they use only surface information.

In recent years, various metrics based on semantic similarity have been proposed as neural network technology has developed. In metrics using pre-trained models (*e.g.* YiSi-1 [6], WE_WPI [7]), various static embedding models (*e.g.* word2vec [8], fastText [9], GloVe [10], BERT [11]) are used to obtain semantic similarities between pairwise word embeddings in references and hypotheses. However, these metrics show difficulty in application to suitable specific domains because they depend on static word embedding models. In metrics using fine-tuned models such as BLEURT [12], COMET [13], and OpenKiwi [14], the estimator models are built by fine-tuned language models using earlier WMT data. Moreover, in metrics using neural models (*e.g.* esim [15], prism [16]), contextual representations that adapt to a specific domain are extracted from the learned model using earlier WMT data. These metrics can accommodate various sentences by fine-tuning of the models. However, they require large amounts of learning data to fine-tune the language models.

As described herein, we propose a new metric based on optimal transport by integration of contextual representations and static word embeddings. The contextual representations are extracted from the neural model (*e.g.* an attention-based LSTM [17]) learned only using source sentences and references, which correspond to the evaluation target hypotheses. Therefore, our proposed metric can adapt to a specific domain without requiring large amounts of learning data such as earlier WMT data. Moreover, in our proposed metrics, the contextual representations are used as the weights for the static word embeddings. This means that the static word embeddings are corrected to adapt the domain in the vector space. Furthermore, the evaluation scores are calculated based on the Earth Mover's Distance (EMD) [18] algorithm, which solves the optimal transport problem, in our proposed metric. As described herein, we call this proposed metric **OPTICS** (automatic metric based on **OP**timal **T**ransport by **I**ntegration of **C**ontextual representations and **S**tatic word embeddings). Through the experiments using WMT 20 metric shared task data, OPTICS outperformed the metric using only the static word embeddings (*i.e.* WE_WPI [7]). Moreover, we confirmed that OPTICS achieves state-of-the-art performance in correlations with human judgments except out-of-English segment-level correlation.

2 Related Work

2.1 Metrics Using Pre-trained Models

YiSi-1 [6] obtains evaluation scores using idf-weighted lexical semantic similarity by the pre-trained embedding models. The idf-weighted lexical semantic similarity is based on cosine similarity between lexical embeddings. In WMT 20 metric shared task, YiSi-1 uses lexical embeddings extracted from pre-trained language models (*e.g.* BERT [11], CamemBERT [19], RoBERTa [20], XLM [21], XLM-RoBERTa [22]). Moreover, it optionally requires a semantic role labeler in the hypotheses for evaluating structural semantic similarity. Also, WE_WPI [7] is a metric based on EMD [18] which solves the problem of the optimal transport. In that case, fastText [9] as static word embedding model is used to calculate similarities between word embeddings in reference and hypothesis. Moreover, it uses the relative difference of word order between a reference and hypothesis as the weight in the calculation of similarity. As a result, it can perform evaluation that specifically examines differences of word order.

2.2 Metrics Using Fine-Tuned Models

BLEURT [12] is a BERT-based regression model. This model is fine-tuned on human ratings from WMT 2015 Metrics to WMT 2019 Metrics. Moreover, it learns to predict widely diverse similarity that includes existing metrics (BERTScore [23] etc.) as mid-training. COMET [13] metrics are built using the Estimator model or the Translation Ranking model. The Estimator model regresses on human judgment such as direct assessment (DA) from WMT 2017 Metrics to WMT 2019 Metrics. The Translation Ranking model is trained to minimize the distance between a better hypothesis and both its corresponding reference and its source sentence. Both models constitute a cross-lingual encoder and a pooling layer. OpenKiwi [14] metrics are Quality Estimation models; they are trained with WMT Metrics data from 2017–2019. OpenKiwi-BERT and OpenKiwi-XLMR respectively use fine-tuning BERT and XLM models in a predictor–estimator architecture.

2.3 Metrics Using Neural Models

Esim [15], a neural model proposed for Natural Language Inference as described in an earlier report [24], has been adapted for MT evaluation. This model encodes a reference and hypothesis with a BiLSTM, and computes attention-weighted representations of each word embedding in the hypothesis with respect to each word embedding in the reference. Moreover, a feedforward regressor over the concatenation of the reference and hypothesis is applied to compute the final predicted score. Esim requires human evaluation data from WMT Metrics as training data. Prism [16] is a multilingual NMT system trained on data for 39 language pairs, with data derived largely from WMT and Wikimatrix, although it requires no human judgment for training. At evaluation time, the model is

used in zero-shot mode to score hypotheses conditioned on their corresponding references.

3 New Automatic Metric: OPTICS

Our proposed metric is constructed from two stages: learning of the encoder for extraction of contextual representations, and calculation of scores based on optimal transport using contextual representations by the trained encoder and the static word embeddings.

3.1 Learning of Encoder for Extracting Contextual Representations

Our proposed metric obtains the encoder for extraction of contextual representations. As described herein, attention-based LSTM, which is a basic encoder–decoder model, is used to obtain the encoder. It remains unclear whether the attention-based LSTM is the best model among various encoder–decoder models, or not. Therefore, this point is an issue that remains for future study. Figure 1 depicts the outline for learning of attention-based LSTM.

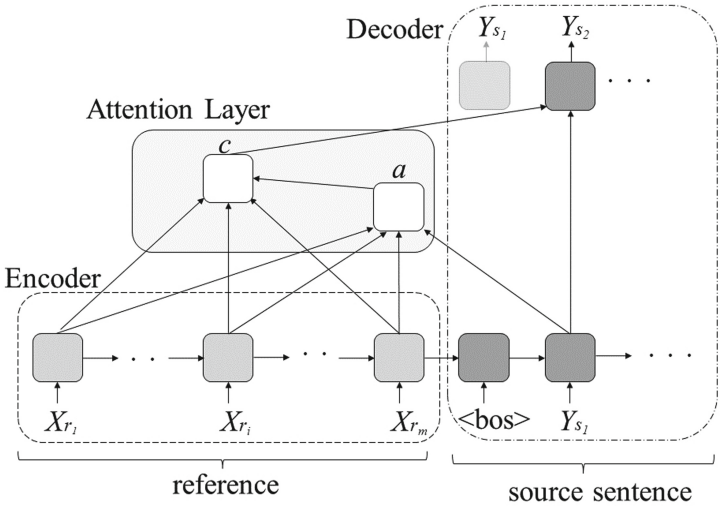


Fig. 1. Outline of learning of attention-based LSTM model.

In Fig. 1, the reference and source sentence corresponding to the evaluation target hypothesis, are used, respectively, as encoder and decoder inputs. As a result, our proposed metric can convert the reference and hypothesis respectively into contextual representations by the encoder of the learned attention-based LSTM model.

3.2 Score Calculation Based on Optimal Transport Using Contextual Representations and Static Word Embeddings

The encoder of the learned neural model is used to obtain the contextual representations of words in the reference and hypothesis. Moreover, those contextual representations are used as features of distributions in EMD, which is an algorithm to solve the optimal transport problem. Figure 2 presents an outline of score calculation.

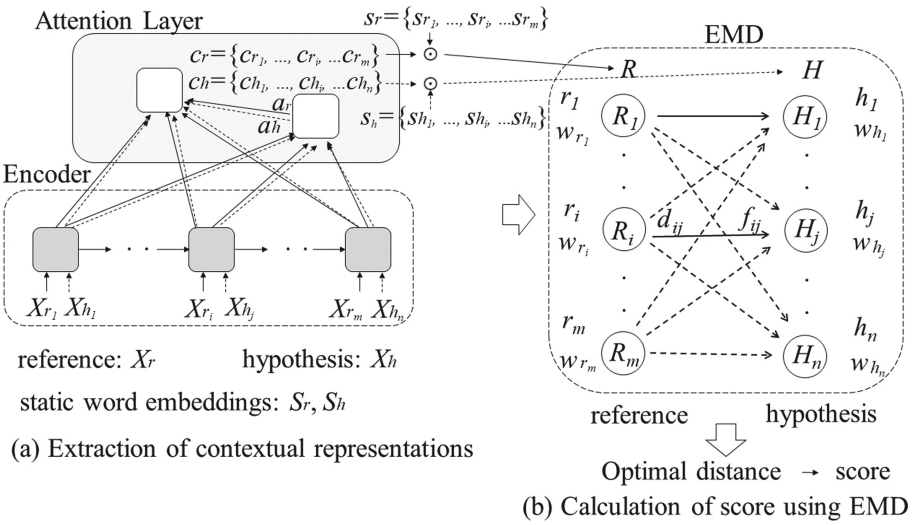


Fig. 2. Outline of score calculation.

In Fig. 2 (a), the contextual representations $c_{r_{1\sim m}}$ of the words in reference $X_{r_{1\sim m}}$ and the contextual representations $c_{h_{1\sim n}}$ of the words in hypothesis $X_{h_{1\sim n}}$ are respectively extracted from the trained encoder; also, m and n indicates the word numbers of reference and hypothesis, respectively. The c_{r_i} and c_{h_j} are respectively integrated into the static word embeddings s_{r_i} and s_{h_j}. The r_i and h_j, which are used as the features in EMD, are obtained as Eqs. (1) and (2).

$$r_{i_k} = exp(\alpha \times c_{r_{i_k}}) \times s_{r_{i_k}} \tag{1}$$

$$h_{j_k} = exp(\alpha \times c_{h_{j_k}}) \times s_{h_{j_k}} \tag{2}$$

In Eqs. (1) and (2), α is a high parameter, and k means each element position of the vector. In Eq. (1), r_{i_k} is calculated using the contextual representation c_{r_i}, which is extracted from the encoder of the attention-based LSTM model, and the static word embedding s_{r_i}. The calculation in Eq. (1) is performed using all elements $\{c_{r_{i_1}}, ..., c_{r_{i_k}}, ..., c_{r_{i_l}}\}$ and $\{s_{r_{i_1}}, ..., s_{r_{i_k}}, ..., s_{r_{i_l}}\}$. In Eq. (2), h_{j_k} is

calculated using the contextual representation of c_{h_j}, which is extracted from the encoder of the attention-based LSTM model, and the static word embedding s_{h_j}. This indicates that the calculation in Eq. (2) is performed using all elements $\{c_{h_{j_1}}, ..., c_{h_{j_k}}, ..., c_{h_{j_l}}\}$ and $\{s_{h_{j_1}}, ..., s_{h_{j_k}}, ..., s_{h_{j_l}}\}$. In Eqs. (1) and (2), the contextual representations c_{r_i} and c_{h_j} are used as the weight for the static word embeddings s_{r_i} and s_{h_j}. Therefore, the static word embeddings are corrected to adapt the domain by using the contextual representations in the vector space.

Word vector r_i in the reference and word vector h_j in the hypothesis are used as EMD features. In Fig. 2 (b), the distributions R and H respectively correspond to the reference and hypothesis. Each signature $\{R_1, ..., R_i, ..., R_m\}$ of distribution R has word vector $\{r_1, ..., r_i, ..., r_m\}$ and weight $\{w_{r_1}, ..., w_{r_i}, ..., w_{r_m}\}$. Each signature $\{H_1, ..., H_j, ..., H_n\}$ of distribution H also has word vector $\{h_1, ..., h_j, ..., h_n\}$ and weight $\{w_{h_1}, ..., w_{h_j}, ..., w_{h_n}\}$. Moreover, m and n respectively means the word numbers of reference and hypothesis. Weights w_{r_i} and w_{h_j} are obtained using sentence-level $tf \cdot idf$. Each weight based on sentence-level $tf \cdot idf$ is defined as Eq. (3).

$$w = tf \times \left(\log \frac{N}{df} + 1.0 \right) \tag{3}$$

In Eq. (3), tf represents the number of times that the word appears in a reference or hypothesis. Also, N denotes the number of all references or all hypotheses, and df stands for the sentence number that the word appears for all references or all hypotheses. Moreover, each weight w_{r_i} and w_{h_j} is respectively defined as Eqs. (4) and (5).

$$\tilde{w}_{r_i} = \frac{w_{r_i}}{\sum_{k=1}^{m} w_{r_k}} \tag{4}$$

$$\tilde{w}_{h_j} = \frac{w_{h_j}}{\sum_{k=1}^{n} w_{h_k}} \tag{5}$$

By Eqs. (4) and (5), the weights are normalized between 0.0 and 1.0. The cosine similarity between two word vectors r_i and h_j is used to calculate the distance d_{ij}. In that case, the penalty based on the difference of word order is applied to cosine similarity. Also, d_{ij} is obtained using the following Eq. (6) as shown below.

$$d_{ij} = \begin{cases} 1.0 - cos_sim(r_i, h_j) \times exp(-pos_inf(X_{r_i}, X_{h_j})) \\ \quad \text{if } cos_sim(r_i, h_j) \text{ is the highest among } cos_sim \text{ between } r_i \text{ and } h_{1 \sim n} \\ 1.0 \quad \text{otherwise} \end{cases} \tag{6}$$

$$pos_inf(X_{r_i}, X_{h_j}) = \left| \frac{pos(X_{r_i})}{m} - \frac{pos(X_{h_j})}{n} \right| \tag{7}$$

In Eq. (6), $cos_sim(r_i, h_j)$ represents cosine similarity between r_i by Eq. (1) and h_j by Eq. (2), and $exp(-pos_inf(X_{r_i}, X_{h_j}))$ represents the penalty to

$cos_sim(r_i, h_j)$ because it is smaller (*i.e.* less than 1.0) as the difference of word order $pos_inf(X_{r_i}, X_{h_j})$ becomes larger. The $pos_inf(X_{r_i}, X_{h_j})$ is obtained by Eq. (7). Here, $pos(X_{r_i})$ and $pos(X_{h_j})$ respectively correspond to word positions in the reference and hypothesis. Our proposed metric can obtain the correct score which specifically emphasizes the difference of word order between the reference and hypothesis. Moreover, in EMD, the overall cost is defined as in Eq. (8).

$$WORK(R, H, F) = \sum_{i=1}^{m} \sum_{j=1}^{n} d_{ij} f_{ij} \tag{8}$$

In Eq. (8), $F = [f_{ij}]$ denotes the total flow; also, EMD obtains F which minimizes the overall cost. Moreover, four constraints are defined for f_{ij} to find minimum F as in the following Eqs. (9)–(12).

$$f_{ij} \geq 0 \qquad 1 \leq i \leq m, \qquad 1 \leq j \leq n \tag{9}$$

$$\sum_{j=1}^{n} f_{ij} \leq w_{r_i} \qquad 1 \leq i \leq m \tag{10}$$

$$\sum_{i=1}^{m} f_{ij} \leq w_{h_j} \qquad 1 \leq j \leq n \tag{11}$$

$$\sum_{i=1}^{m} \sum_{j=1}^{n} f_{ij} = min\left(\sum_{i=1}^{m} w_{r_i}, \sum_{j=1}^{n} w_{h_j} \right) \tag{12}$$

In fact, Eq. (9) shows that "supplies" must be transferred from R to H, not from H to R. Also, Eq. (10) shows that supplies exceeding the weights of R cannot be transferred to H; Eq. (11) shows that supplies exceeding the weights of H cannot be received from R. Moreover, Eq. (12) means that the upper limit of the total supply corresponds to the smaller of the total weight of R and the total weight of H. Also, EMD is defined as shown below.

$$EMD(R, H) = \frac{min(WORK(R, H, F))}{\sum_{i=1}^{m} \sum_{j=1}^{n} f_{ij}} \tag{13}$$

The EMD in Eq. (13) is normalized from 0.0 to 1.0. The value approaches 0.0 as the distance between the two distributions becomes smaller. In our proposed metric, the final score is defined as Eq. (14).

$$score = 1.0 - EMD(R, H) \tag{14}$$

Therefore, the score value in Eq. (14) is closer to 1.0 as the evaluation is higher.

4 Experiments

4.1 Experiment Data and Procedure

We used WMT20 metrics shared task [25] data for experiments. For learning the attention-based LSTM in our proposed metric, we copied the sentence pair, which is the source sentence and its reference, ten times to increase the learning data for each language pair. In that case, we used only source sentences and their references, which correspond to the evaluation target hypotheses in WMT20 metrics shared task data. The maximum number of pairs of source sentences and its references among all language pairs is 29,710, which is the number of pairs of both Inuktitut–English and English–Inuktitut. On the other hand, the minimum number of pairs is 7,850 which is the number of pairs of German–English. Using those learning data, we produced the attention-based LSTM model in Fig. 1 to extract the contextual representations in each language pair.

In Fig. 2, the numbers of dimensions r_i, h_j, c_{r_i}, c_{h_j}, s_{r_i}, and s_{h_j} respectively denote 300. The number of epochs is 80 in each language pair. The contextual representations of the reference c_{r_i} and those of the hypothesis c_{h_j} are extracted, respectively, from the encoder of the learned attention-based LSTM models. Moreover, 0.1 was used as the high parameter α in Eqs. (1) and (2) by preliminary experimentation. Our proposed metric (*i.e.* OPTICS) and WE_WPI [7] used fastText [9] as the static word embedding models for reference and hypothesis. In OPTICS and WE_WPI, the averages of the segment-level scores are used as system-level scores.

Next, we obtained the correlation coefficients with human judgments to evaluate OPTICS and the state-of-the-art metrics using the script provided by WMT20. The script is provided at http://www.statmt.org/wmt20/results.html.

In WE_WPI and OPTICS, fastText model of Inuktitut is not provided at https://fasttext.cc/docs/en/crawl-vectors.html. Therefore, correlations of English–Inuktitut in WE_WPI were not obtained. In OPTICS, only contextual representations, which are extracted from the encoder of the neural model learned using the pairs of English source sentence and Inuktitut reference, were used. In that case, the vectors, of which all elements are 1.0, were used as the static word embeddings (*i.e.* $s_{r_{i_k}}$, $s_{h_{j_k}}$) of Inuktitut.

4.2 Experiment Results and Discussion

Table 1. Pearson correlations of to-English at the system-level.

	cs-en	de-en	ja-en	pl-en	ru-en	ta-en	zh-en	iu-en	km-en	ps-en	Avg.
BLEURT -extend	**0.771**	0.985	**0.961**	**0.551**	0.900	**0.897**	0.945	0.789	0.985	**0.942**	0.873
COMET	**0.783**	**0.998**	0.964	**0.591**	**0.923**	0.880	**0.952**	0.852	0.971	**0.941**	0.886
esim	**0.790**	**0.998**	**0.983**	**0.591**	**0.928**	0.885	**0.963**	0.807	0.929	**0.929**	0.8803
OpenKiwi -Bert	0.726	0.989	0.735	0.355	**0.862**	0.645	0.625	−0.126	0.751	0.753	0.632
OpenKiwi -XLMR	**0.760**	0.995	0.931	0.442	0.859	0.792	0.905	0.271	0.880	0.865	0.770
prism	**0.818**	**0.998**	**0.974**	**0.502**	**0.908**	0.898	**0.957**	0.833	0.950	**0.966**	0.8804
YiSi-1	**0.832**	**0.998**	**0.982**	**0.543**	**0.915**	**0.925**	**0.961**	0.834	0.977	**0.953**	0.892
WE_WPI (Baseline)	**0.838**	**0.998**	**0.973**	**0.573**	**0.939**	**0.933**	**0.965**	0.776	**0.993**	**0.922**	0.891
OPTICS (This work)	**0.849**	**0.998**	**0.972**	**0.541**	**0.934**	**0.933**	0.964	0.824	**0.993**	**0.929**	0.894

Table 2. Pearson correlations of out-of-English at the system-level.

	en-cs	en-de	en-ja	en-pl	en-ru	en-ta	en-zh	en-iu _fill	en-iu _news	Avg.
BLEURT-extend	0.989	**0.969**	0.944	**0.982**	**0.980**	0.940	0.928	0.823	0.762	0.924
COMET	0.978	**0.972**	0.974	**0.981**	0.925	0.944	0.007	0.860	**0.858**	0.833
esim	0.908	**0.979**	0.993	**0.969**	**0.967**	0.937	**0.972**	0.814	0.760	0.922
OpenKiwi-Bert	0.920	0.852	0.363	0.903	0.834	0.846	0.551	0.573	0.808	0.739
OpenKiwi-XLMR	0.972	**0.968**	0.992	0.957	0.875	0.910	−0.010	0.513	0.680	0.762
prism	0.949	0.958	0.932	0.958	0.724	0.863	0.221	**0.957**	**0.945**	0.834
YiSi-1	0.922	**0.971**	0.969	**0.964**	**0.926**	0.973	0.959	0.554	0.523	0.862
WE_WPI (Baseline)	0.879	0.941	0.964	0.894	**0.945**	0.936	0.911	–	–	–
OPTICS (This work)	0.868	0.944	0.964	0.913	**0.952**	0.935	0.910	0.586	0.695	0.863

Table 3. Kendall correlations of to-English at the segment-level.

	cs-en	de-en	iu-en	ja-en	km-en	pl-en	ps-en	ru-en	ta-en	zh-en	Avg.
BLEURT -extend	0.127	0.448	0.259	0.271	0.330	0.044	0.161	0.101	0.246	0.137	0.212
COMET	0.129	0.485	0.281	0.274	0.298	0.099	0.158	0.156	0.241	0.171	0.229
esim	0.110	0.454	0.241	0.239	0.300	0.058	0.147	0.084	0.208	0.138	0.198
OpenKiwi -Bert	0.036	0.379	−0.005	0.110	0.168	−0.033	0.076	−0.033	0.118	0.029	0.085
OpenKiwi -XLMR	0.093	0.463	0.056	0.220	0.244	0.059	0.106	0.092	0.188	0.115	0.164
prism	0.143	0.475	0.255	0.272	0.304	0.109	0.165	0.145	0.237	0.167	0.227
YiSi-1	0.117	0.468	0.253	0.277	0.316	0.042	0.147	0.091	0.248	0.146	0.211
WE_WPI (Baseline)	0.102	0.474	0.218	0.238	0.239	0.080	0.134	0.133	0.222	0.151	0.199
OPTICS (This work)	0.103	0.475	0.231	0.233	0.235	0.089	0.149	0.134	0.222	0.154	0.203

Tables 1 and 2 respectively present Pearson correlations of to-English and out-of-English system-level metrics with direct assessment (DA) human assessment over MT systems using the *newstest2020* references. Tables 3 and 4 respectively present segment-level metric results for to-English and out-of-English language pairs. These coefficients are Kendall's Tau formulations of the segment-level metric. The highlighted values shown in bold typeface in Tables 1–2 denote that the correlations of metrics are not significantly outperformed by any other for that language pair. "Avg." stands for the average of correlations of all language pairs.

Table 4. Kendall correlations of out-of-English in the segment-level.

	en-cs	**en-de**	**en-iu**	**en-ja**	**en-pl**	**en-ru**	**en-ta**	**en-zh**	**Avg.**
BLEURT-extend	0.689	0.447	0.359	0.533	0.430	0.305	0.643	0.460	0.483
COMET	0.668	0.468	0.322	0.624	0.462	0.344	0.671	0.432	0.499
esim	0.469	0.347	0.122	0.522	0.312	0.224	0.599	0.391	0.373
OpenKiwi-Bert	0.262	0.168	−0.115	−0.529	0.153	0.164	0.169	0.077	0.044
OpenKiwi-XLMR	0.607	0.369	0.060	0.553	0.347	0.279	0.604	0.377	0.400
prism	0.619	0.447	0.452	0.579	0.414	0.283	0.448	0.397	0.455
YiSi-1	0.550	0.427	0.251	0.568	0.349	0.256	0.669	0.463	0.442
WE_WPI (Baseline)	0.477	0.331	–	0.502	0.276	0.192	0.376	0.379	–
OPTICS (This work)	0.471	0.339	0.250	0.500	0.280	0.184	0.367	0.381	0.347

The "Avg." of correlations with human judgment in Tables 1 and 3 expresses that OPTICS outperforms WE_WPI, which is the metric based on optimal transport using static word embeddings only. Moreover, OPTICS achieves state-of-the-art performance except out-of-English segment-level correlations of Table 4. The difference between "Avg." of top rank (*i.e.* COMET) and that of OPTICS is not small in out-of-English segment-level correlations of Table 4 because the correlation of "**en-ta**" in OPTICS is low. To ascertain the cause, it is necessary to investigate the related details. On the other hand, the correlation of "**en-iu**" of OPTICS in Table 4 is not a severe difficulty compared with that of "**en-ta**" of OPTICS, although OPTICS calculated the scores only using the contextual representations in **"en-iu"**. Therefore, this result represents the effectiveness of the contextual representations in OPTICS.

5 Conclusion

As described in this paper, we proposed OPTICS as a new automatic metric based on optimal transport by integration of contextual representations and static word embeddings. In OPTICS, after the contextual representations

are extracted from the encoder of the learned neural models, they are used as weights for static word embeddings. Through the experimentally obtained results, we confirmed that OPTICS can obtain high correlations with human judgment to WE_WPI, which uses only static word embeddings. Moreover, OPTICS achieved state-of-the-art performance, although the correlations of out-of-English segment-level are insufficient.

Future work will be undertaken to investigate the effects of other neural models (*e.g.* Transformer [26]), which extract contextual representations. In that case, the investigation about the epoch number and the suitable learning data is also required. Moreover, we expect to perform a meta-evaluation using various data (*e.g.* WMT 21 data using MQM-based human evaluation [27]).

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